

CSC 470 — SOFTWARE ENGINEERING LAB

AI Resume Analyzer Platform

User Requirements • System Requirements • Use Case Diagram • Non-Functional Requirements

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User Requirements

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Authentication and Session Management

U

Resume Upload and Processing

F

AI Resume Analysis and Feedback

D

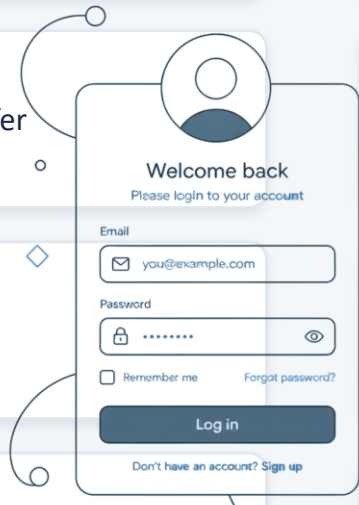
User Dashboard and Analytics

M

Admin Panel Management



Authentication and Session Management

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- 1.1 Sign up with name, email, password & confirm password
 - 1.2 Show "Passwords do not match" instantly if they differ
 - 1.3 Users & admins log in with email and password
 - 1.4 Login creates a secure session storing the user's role
 - 1.5 Show "Invalid email or password" for wrong login info
 - 1.6 System checks the user's role right after login
 - 1.7 Admins are sent to the Admin Dashboard
 - 1.8 Users are sent to the Resume Analysis Dashboard
 - 1.9 Logout ends the session and returns to the login page

Resume Upload and Processing

2.1 Upload by drag-and-drop or by choosing a file

2.2 Accepts PDF, Word, or Text files, up to 5 MB

2.3 Shows an error right away if the file is invalid

2.4 Shows a live resume preview with no page reload

2.5 Extracts resume text using a parsing tool

2.6 "Analyze Resume" click prepares text for AI

AI Scoring and Feedback Engine

3.1 Users add a Job Description with the resume

3.2 Resume & job description are sent to the AI

3.3 Shows a loading message while AI is analyzing

3.4 AI generates an ATS Match Score from 0% to 100%

3.5 Feedback covers Skills, Content, Structure & Writing Style

3.6 System finds important missing skills or keywords

3.7 Missing keywords are shown as recommendation tags

3.8 Users can rate the analysis once it's complete

User Dashboard and Analytics

4.1 View all past resumes, ATS scores & analysis dates

4.2 A "Score Over Time" chart shows progress

4.3 Compare a past resume and job description side by side



Admin Panel Management

5.1 Admins get a separate Admin Dashboard after login

5.2 Shows users, resumes checked, avg score & active sessions

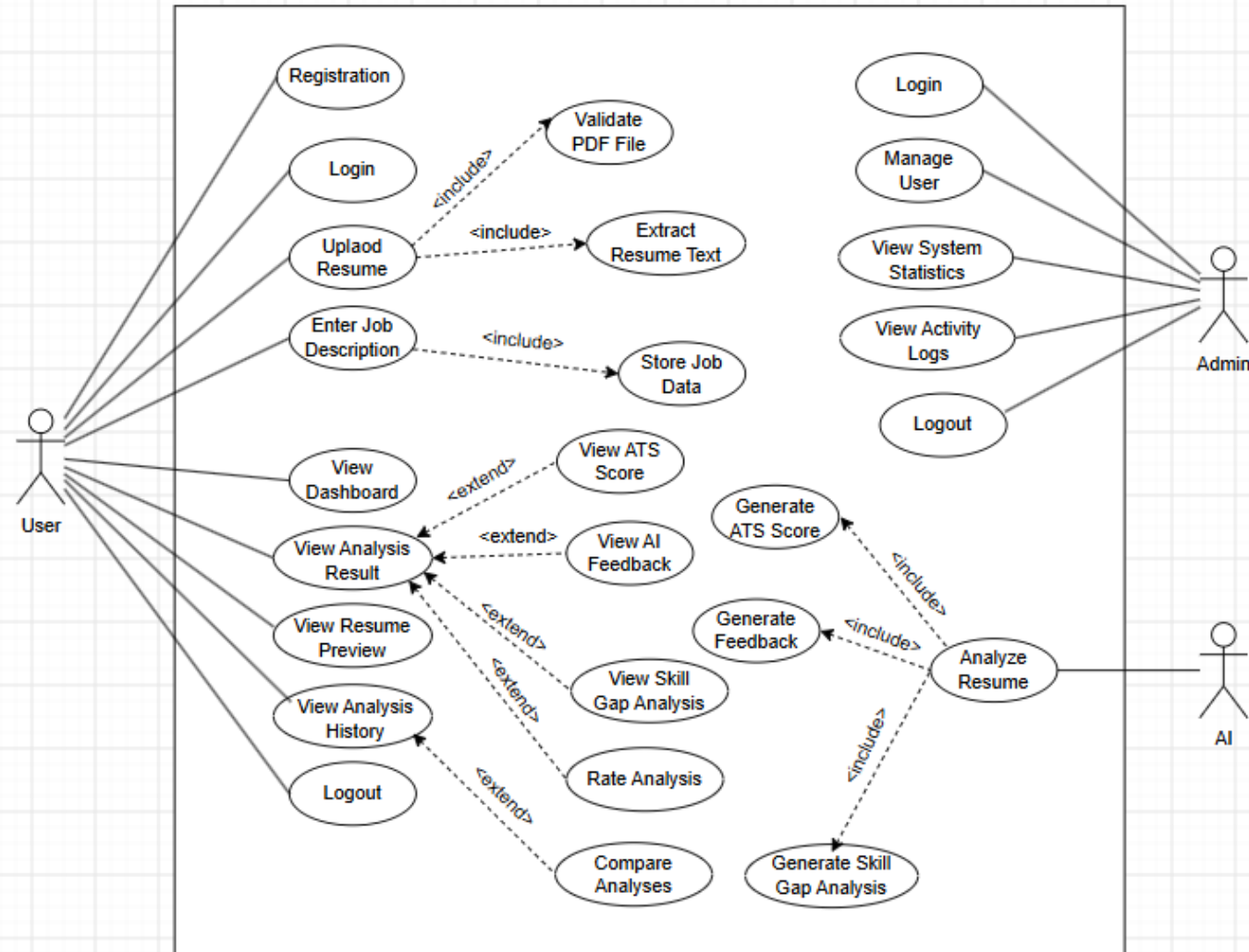
5.3 Searchable, paginated list of all users & their roles

5.4 Change a user's status between Active and Banned

5.5 View activity logs, AI status, and error logs



Use Case Diagram



Non-Functional Requirements

NFR-1

Performance

Fast uploads; AI replies in about 15 seconds

NFR-2

Security

Hashed passwords; only admins reach admin pages

NFR-3

Reliability

Accurate text extraction; no data loss on errors

NFR-4

Usability

Clean dark theme; simple, clear navigation

NFR-5

Maintainability

Layered, well-documented code; easy to extend

NFR-6

Scalability

Handles growth in users, resumes & AI requests

NFR-7

Availability

Always on for upload, analysis & review

NFR-8

Data Integrity

Accurate, consistent links between records



Thank You

Questions & Discussion Welcome